

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Modelling with Sequences & Series

*Modelling questions turn real words into sequence variables
and then interpret the result.*

MODELLING WITH SEQUENCES & SERIES · P2 · 08

ELITEIGCSE.COM · WHATSAPP +20 112 000 9622



DR ESLAM AHMED · CAIRO UNIVERSITY FACULTY OF ENGINEERING

The Map

This strategy booklet was mined from the local topic problem/answer PDFs, Dr Eslam's source notes, and the WMA12 website answer bank. The topic reduces to these exam moves, with hard variants kept visible on each page.

01 MODEL: Define The Model

TRIGGER *context with repeated change*

02 FIT: Fit Unknowns

TRIGGER *two pieces of data*

03 LIMIT: Long-Term Limit

TRIGGER *maximum total, limiting value*

04 INTERP: Interpret The Answer

TRIGGER *least n , first time, exceeds*

BANK EVIDENCE Local pages: 29 problem, 63 answer, 4 notes. Website primary entries: 10.

01 MODEL: Define The Model

TRIGGER *context with repeated change*

BECOMES Name the first value and the repeated change.

FIRST LINE TO WRITE

$$u_n = a + (n - 1)d \quad \text{or} \quad u_n = ar^{n-1}$$

SIMPLEST STRATEGY

- 1 **Mark** the first term.
- 2 **Observe** add or multiply change.
- 3 **Define** n .
- 4 **Enter** the right formula.
- 5 **Label** units.

WORKED MODEL

| A salary increasing by 200 each year is arithmetic.

HARD VARIANTS

- 1 Check whether the first payment is at the start or end of the first period.
- 2 Use arithmetic for fixed increase and geometric for percentage change.
- 3 Keep units with money, population, height, or area.

— BOTTOM LINE

A model is only useful if n and units are defined.

02 FIT: Fit Unknowns

TRIGGER *two pieces of data*

BECOMES Use the data to find model constants.

FIRST LINE TO WRITE

| data point $\Rightarrow$ equation

SIMPLEST STRATEGY

- 1 Find the two facts.
- 2 Insert each into the model.
- 3 Turn them into constants.

WORKED MODEL

| $u_1 = 500, u_4 = 650 \Rightarrow d = 50.$

HARD VARIANTS

- 1 If data years are not consecutive, count the number of intervals.
- 2 Percentage decrease means $r < 1$.
- 3 Check the model reproduces both given values.

— BOTTOM LINE

Data points determine the model.

03 LIMIT: Long-Term Limit

TRIGGER *maximum total, limiting value*

BECOMES Use sum to infinity or long-term behaviour.

FIRST LINE TO WRITE

$$S_{\infty} = \frac{a}{1-r} \quad (|r| < 1)$$

SIMPLEST STRATEGY

- 1 Look for repeated percentage.
- 2 Inspect $|r| < 1$.
- 3 Make the limiting formula.
- 4 Interpret the value.
- 5 Tell units.

WORKED MODEL

| A bounce retaining $\frac{3}{4}$ of height has total limiting distance from a geometric series.

HARD VARIANTS

- 1 For decay, decide whether the first term is included in the total.
- 2 If a threshold is asked, solve an inequality, not just a limit.
- 3 Round according to the real context.

— BOTTOM LINE

Limits need interpretation, not just calculation.

04 INTERP: Interpret The Answer

TRIGGER *least n, first time, exceeds*

BECOMES Turn algebraic solution into context language.

FIRST LINE TO WRITE

$$u_n > \text{target} \quad \text{or} \quad S_n > \text{target}$$

SIMPLEST STRATEGY

- 1 Identify what must exceed or fall below.
- 2 Navigate the inequality.
- 3 Take integer n .
- 4 Explain in context.
- 5 Round in the correct direction.
- 6 Present units.

WORKED MODEL

| If $n > 6.2$, the first whole period is $n = 7$.

HARD VARIANTS

- 1 For 'at least' or 'exceeds', test the boundary integer.
- 2 If the model starts at $n = 0$, translate n into calendar time.
- 3 Give a sentence, not only a number.

— BOTTOM LINE

Context answers often need rounding up.

Quick Reference

TRIGGER → FIRST ACTION

TRIGGER	FIRST ACTION
context with repeated change	$u_n = a + (n - 1)d$ or $u_n = ar^{n-1}$
two pieces of data	data point $\Rightarrow$ equation
maximum total, limiting value	$S_\infty = \frac{a}{1 - r}$ ($ r < 1$)
least n , first time, exceeds	$u_n > \text{target}$ or $S_n > \text{target}$